

Fórmula para el laplaciano en función de las derivadas de Wirtinger

Lema 1. Sea $f \in \mathcal{C}^2(\mathbb{D})$, entonces $\forall z \in \mathbb{D} : \Delta f(z) = 4 \frac{\partial^2 f}{\partial z \partial \bar{z}}(z) = 4 \frac{\partial^2 f}{\partial \bar{z} \partial z}(z)$.

Demostración: Desarrollando las derivadas de Wirtinger en el lado derecho, tenemos:

$$\begin{aligned} \frac{\partial^2 f}{\partial z \partial \bar{z}} &= \frac{\partial}{\partial z} \left(\frac{\partial f}{\partial \bar{z}} \right) = \frac{\partial}{\partial z} \left(\frac{1}{2} \left(\frac{\partial f}{\partial x} + i \frac{\partial f}{\partial y} \right) \right) \\ &= \frac{1}{2} \left[\frac{1}{2} \left(\frac{\partial^2 f}{\partial x^2} - i \frac{\partial^2 f}{\partial y \partial x} \right) + \frac{1}{2} i \left(\frac{\partial^2 f}{\partial x \partial y} - i \frac{\partial^2 f}{\partial y^2} \right) \right] \\ &= \frac{1}{4} \left(\frac{\partial^2 f}{\partial x^2} - i \frac{\partial^2 f}{\partial y \partial x} + i \frac{\partial^2 f}{\partial x \partial y} + \frac{\partial^2 f}{\partial y^2} \right) = \frac{1}{4} \left(\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} \right) = \frac{1}{4} \Delta f. \end{aligned}$$

Por tanto, $\Delta = 4 \frac{\partial^2}{\partial z \partial \bar{z}}$ (también se puede ver que $\Delta = 4 \frac{\partial^2}{\partial \bar{z} \partial z}$ de forma análoga). ■